

HEESEONG SHIN

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RESEARCH INTEREST

Vision-Language Models, Semantic Segmentation, Diffusion Models, 3D Vision

EDUCATION

KAIST

M.S. & Ph.D in Graduate School of Artificial Intelligence
Advisor: Seungryong Kim

Sep. 2024 - Present
Seoul, Korea

Korea University

M.S. in Computer Science and Engineering
Advisor: Seungryong Kim

Mar. 2023 - Aug. 2024
Seoul, Korea

Korea University

B.S. in Computer Science and Engineering

Mar. 2017 - Feb. 2023
Seoul, Korea

WORK EXPERIENCE

NAVER AI Lab

Research Intern
Mentor: Taekyung Kim

Dec. 2024 - Jun. 2025
Seongnam, Korea

PUBLICATION

Heeseong Shin, Byeongho Heo, Dongyoon Han, Seungryong Kim and Taekyung Kim
“Briding Stable Diffusion to Control via Learnable Condition,”
Under Review.

Heeji Yoon*, Jaewoo Jung*, Junwan Kim*, Hyungyu Choi, **Heeseong Shin**, Sangbeom Lim, Honggyu An, Chaehyun Kim, Jisang Han, Donghyun Kim, Chanho Eom, Sunghwan Hong and Seungryong Kim
“VIRAL: Visual Representation Alignment for Multimodal Large Language Models,”
Under Review.

Chaehyun Kim, **Heeseong Shin**, Eunbin Hong, Heeji Yoon, Anurag Arnab, Paul Hongsuck Seo, Sunghwan Hong and Seungryong Kim
“Seg4Diff: Unveiling Open-Vocabulary Segmentation in Text-to-Image Diffusion Transformers,”
Conference on Neural Information Processing Systems (NeurIPS), San Diego, 2025.

Heeji Yoon*, **Heeseong Shin***, Eunbin Hong, Hyunwook Choi, Daun Jeong, Hansang Cho, and Seungryong Kim (*: Equal Contribution)
“**S⁴M**: Boosting Semi-Supervised Instance Segmentation with SAM,”
IEEE International Conference on Computer Vision (ICCV), Honolulu, 2025.

Sunghwan Hong*, Jaewoo Jung*, **Heeseong Shin**, Jisang Han, Jiaolong Yang, Seungryong Kim, and Chong Luo.
“PF3plat: Pose-Free Feed-Forward 3D Gaussian Splatting,”
International Conference on Machine Learning (ICML), Vancouver, 2025.

Heeseong Shin, Chaehyun Kim, Sunghwan Hong, Seokju Cho, Anurag Arnab[†], Paul Hongsuck Seo[†], and Seungryong Kim[†]
“Towards Open-Vocabulary Semantic Segmentation Without Semantic Labels,”
Conference on Neural Information Processing Systems (NeurIPS), Vancouver, 2024.

Susung Hong, Junyoung Seo, **Heeseong Shin**, Sunghwan Hong, and Seungryong Kim.
“Large Language Models are Frame-level Directors for Zero-shot Text-to-Video Generation,”
First Workshop on Controllable Video Generation at ICML (ICMLW), Vienna, 2024.

Sunghwan Hong, Jaewoo Jung, **Heeseong Shin**, Jiaolong Yang, Seungryong Kim, and Chong Luo.
“Unifying Correspondence, Pose and NeRF for Pose-Free Novel View Synthesis from Stereo Pairs,”
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, 2024. **Highlight**

Seokju Cho*, **Heeseong Shin***, Sunghwan Hong, Anurag Arnab, Paul Hongsuck Seo[†], and Seungryong Kim[†] (*: Equal Contribution)
“CAT-Seg: Cost Aggregation for Open-Vocabulary Semantic Segmentation,”
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, 2024. **Highlight**

ACADEMIC SERVICES

Reviewer

- CVPR - 2024, 2025, 2026
- NeurIPS - 2024, 2025
- AAAI - 2025

PROJECT EXPERIENCE

Samsung Electro-Mechanics Jul. 2023 - Jun. 2024
Development on semi-supervised model for MLCC anomaly segmentation

Samsung Electronics Jan. 2023 - Jun. 2023
Development on Transformer-based model for de-mosaicing camera sensor images.

TEACHING EXPERIENCE

Korea University Spring 2024
Teaching Assistant, DATA302: Introduction to Computer Vision and Applications

Samsung Electronics Sep. 2023
Teaching Assistant, Computer Vision Learning Course

Samsung Electronics Sep. 2023 - Nov. 2023
Teaching Assistant, Multimodal Learning

Samsung Electronics Sep. 2024 - Nov. 2024
Teaching Assistant, Multimodal Learning

PERSONAL EXPERIENCE

Republic of Korea Army May 2019 - Dec. 2020
Fulfilled mandatory military service as Intelligence Operations Sergeant

PERSONAL AND TECHNICAL SKILLS

Languages	Korean, English
DL, ML	PyTorch, Numpy
Programming	Python, Java, C/C++